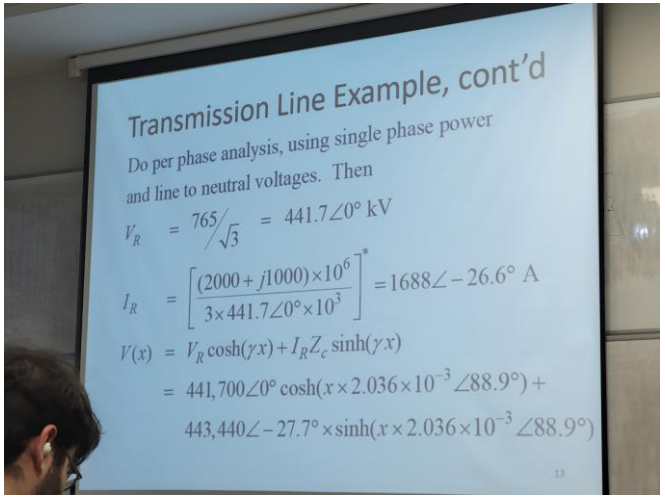
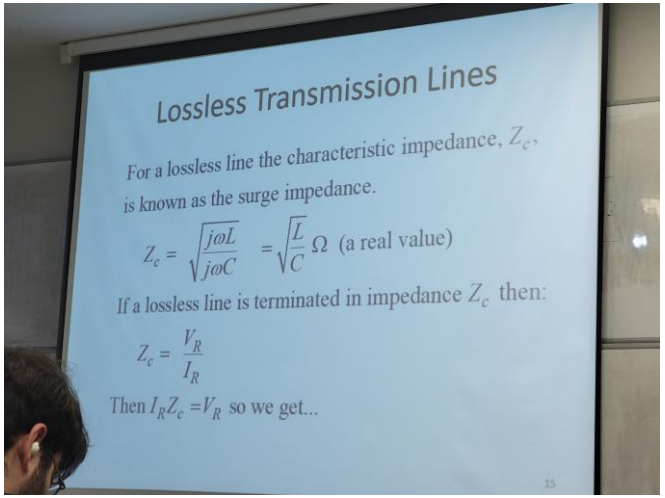
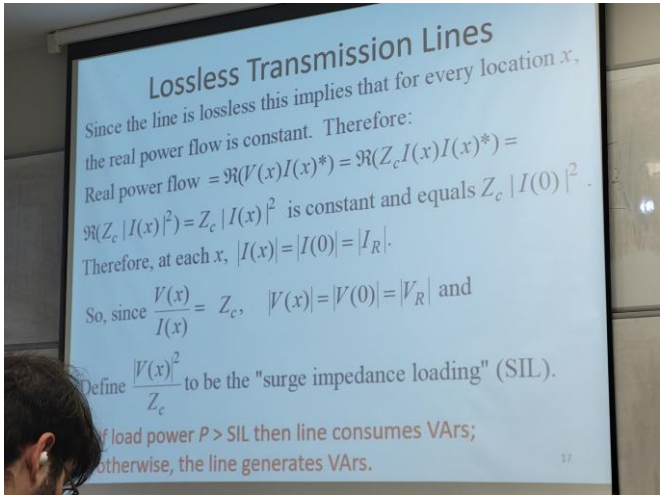
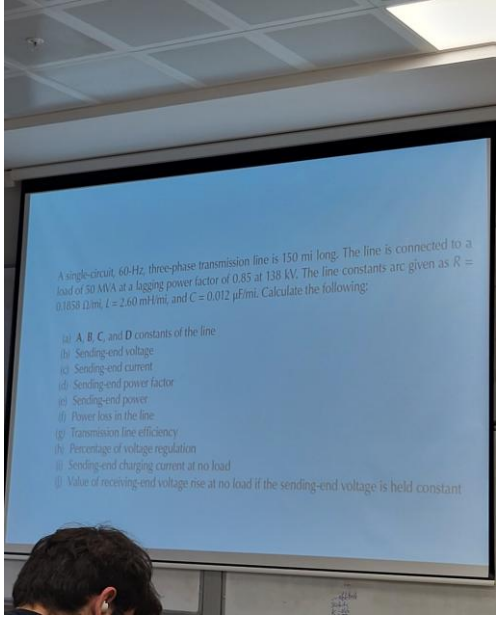
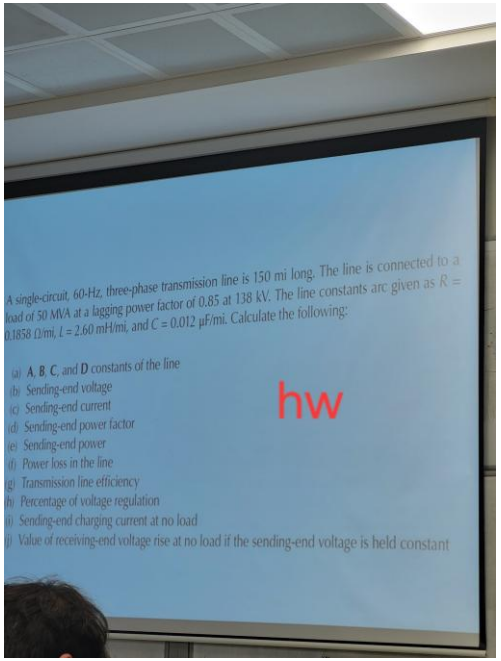
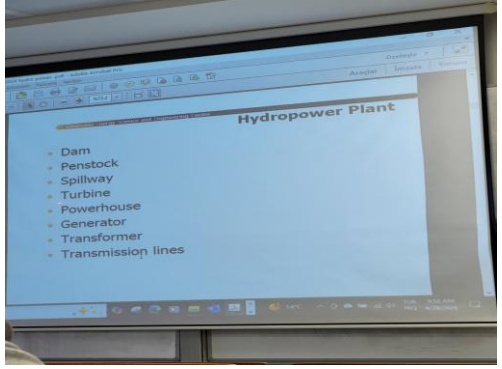

Transmission Line Example, cont'd
Do per phase analysis, using single phase power and line to neutral voltages. Then
$V_R = 765/\sqrt{3} = 441.7\angle 0°$ kV
$I_R = \left[\frac{(2000+j1000)\times 10^6}{3\times 441.7\angle 0°\times 10^3}\right]^* = 1688\angle -26.6°$ A
$V(x) = V_R\cosh(\gamma x) + I_R Z_c \sinh(\gamma x)$
$= 441{,}700\angle 0°\cosh(x\times 2.036\times 10^{-3}\angle 88.9°) +$
$443{,}440\angle -27.7°\times\sinh(x\times 2.036\times 10^{-3}\angle 88.9°)$
13

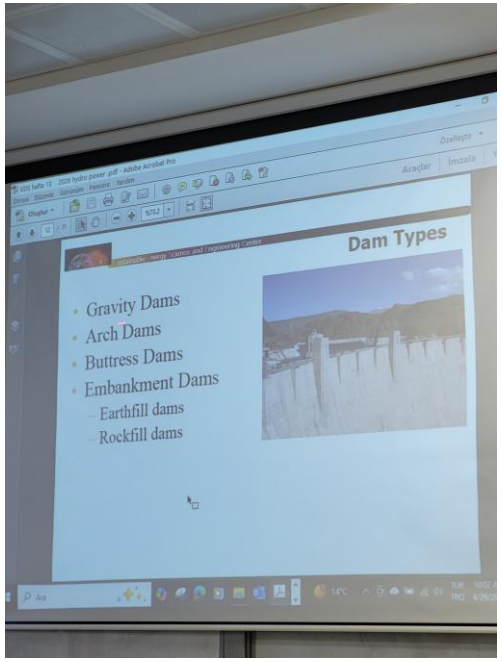

Lossless Transmission Lines
For a lossless line the characteristic impedance, Z_c, is known as the surge impedance.
$Z_c = \sqrt{\frac{j\omega L}{j\omega C}} = \sqrt{\frac{L}{C}}\ \Omega$ (a real value)
If a lossless line is terminated in impedance Z_c then:
$Z_c = \frac{V_R}{I_R}$
Then $I_R Z_c = V_R$ so we get...
15

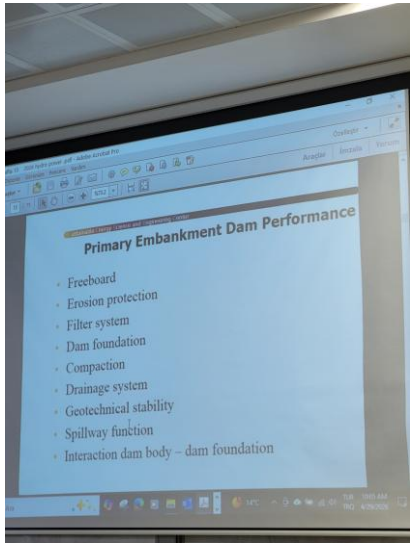

Lossless Transmission Lines
Since the line is lossless this implies that for every location x, the real power flow is constant. Therefore:
Real power flow $= \Re(V(x)I(x)^*) = \Re(Z_c I(x)I(x)^*) =$
$\Re(Z_c|I(x)|^2) = Z_c|I(x)|^2$ is constant and equals $Z_c|I(0)|^2$.
Therefore, at each x, $|I(x)| = |I(0)| = |I_R|$.
So, since $\frac{V(x)}{I(x)} = Z_c$, $|V(x)| = |V(0)| = |V_R|$ and
Define $\frac{|V(x)|^2}{Z_c}$ to be the "surge impedance loading" (SIL).
f load power P > SIL then line consumes VArs;
otherwise, the line generates VArs.
17

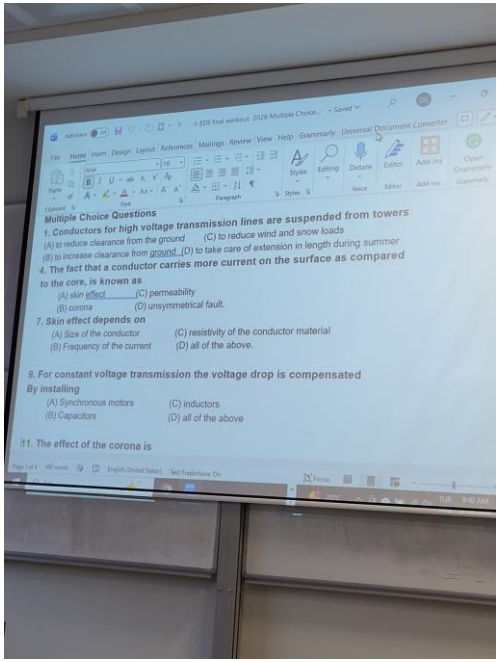

A single-circuit, 60-Hz, three-phase transmission line is 150 mi long. The line is connected to a load of 50 MVA at a lagging power factor of 0.85 at 138 kV. The line constants are given as R = 0.1858 Ω/mi, L = 2.60 mH/mi, and C = 0.012 μF/mi. Calculate the following:
(a) A, B, C, and D constants of the line
(b) Sending-end voltage
(c) Sending-end current
(d) Sending-end power factor
(e) Sending-end power
(f) Power loss in the line
(g) Transmission line efficiency
(h) Percentage of voltage regulation
(i) Sending-end charging current at no load
(j) Value of receiving-end voltage rise at no load if the sending-end voltage is held constant

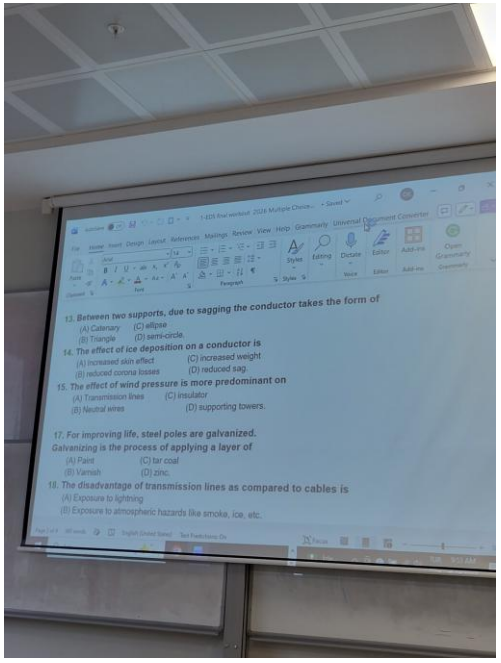

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hw

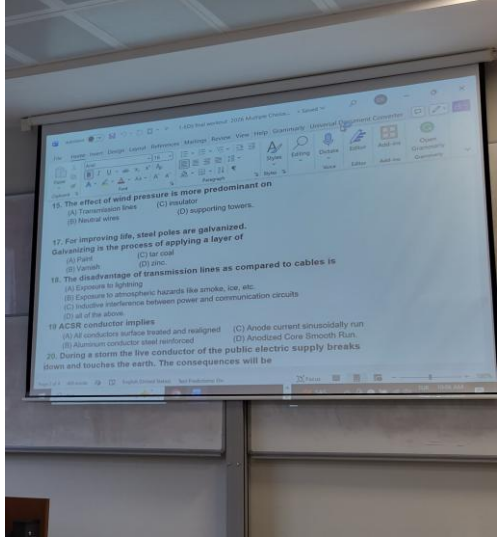

Hydropower Plant
Dam
Penstock
Spillway
Turbine
Powerhouse
Generator
Transformer
Transmission lines

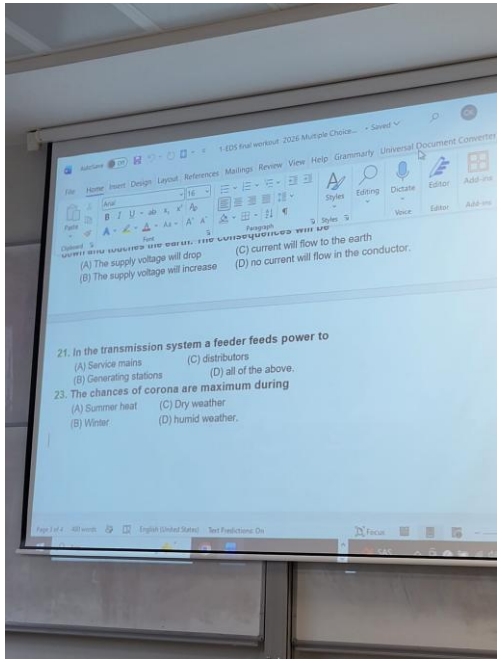

Dam Types
Gravity Dams
Arch Dams
Buttress Dams
Embankment Dams
Earthfill dams
Rockfill dams

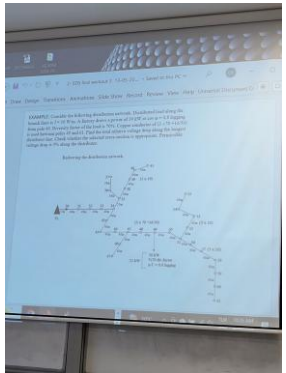

Primary Embankment Dam Performance
Freeboard
Erosion protection
Filter system
Dam foundation
Compaction
Drainage system
Geotechnical stability
Spillway function
Interaction dam body – dam foundation

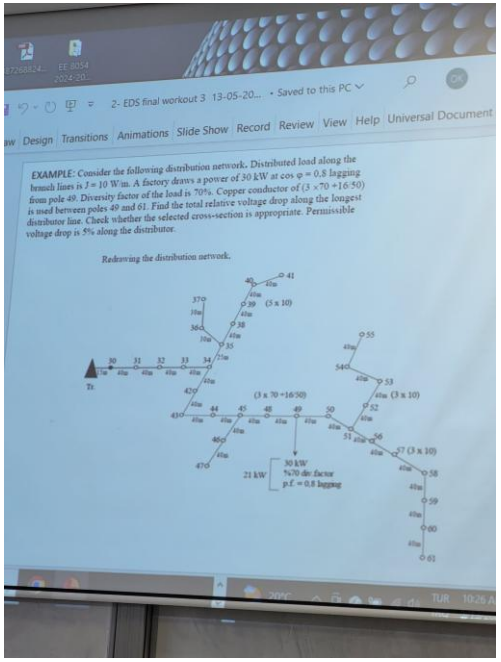

Multiple Choice Questions
1. Conductors for high voltage transmission lines are suspended from towers
(A) to reduce clearance from the ground (C) to reduce wind and snow loads
(B) to increase clearance from ground (D) to take care of extension in length during summer
4. The fact that a conductor carries more current on the surface as compared to the core, is known as
(A) skin effect (C) permeability
(B) corona (D) unsymmetrical fault.
7. Skin effect depends on
(A) Size of the conductor (C) resistivity of the conductor material
(B) Frequency of the current (D) all of the above.
9. For constant voltage transmission the voltage drop is compensated By installing
(A) Synchronous motors (C) inductors
(B) Capacitors (D) all of the above
11. The effect of the corona is

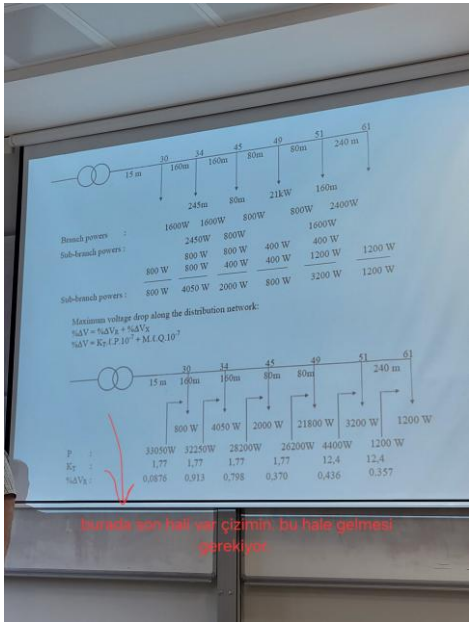

13. Between two supports, due to sagging the conductor takes the form of
(A) Catenary (C) ellipse
(B) Triangle (D) semi-circle.
14. The effect of ice deposition on a conductor is
(A) increased skin effect (C) increased weight
(B) reduced corona losses (D) reduced sag.
15. The effect of wind pressure is more predominant on
(A) Transmission lines (C) insulator
(B) Neutral wires (D) supporting towers.
17. For improving life, steel poles are galvanized. Galvanizing is the process of applying a layer of
(A) Paint (C) tar coat
(B) Varnish (D) zinc.
18. The disadvantage of transmission lines as compared to cables is
(A) Exposure to lightning
(B) Exposure to atmospheric hazards like smoke, ice, etc.

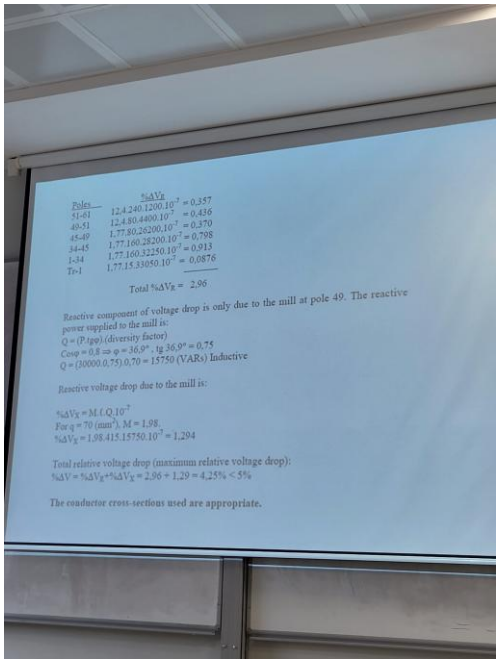

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17. For improving life, steel poles are galvanized.
Galvanizing is the process of applying a layer of
(A) Paint (C) tar coal
(B) Varnish (D) zinc.
18. The disadvantage of transmission lines as compared to cables is
(A) Exposure to lightning
(B) Exposure to atmospheric hazards like smoke, ice, etc.
(C) Inductive interference between power and communication circuits
(D) all of the above.
19 ACSR conductor implies
(A) All conductors surface treated and realigned (C) Anode current sinusoidally run
(B) Aluminium conductor steel reinforced (D) Anodized Core Smooth Run.
20. During a storm the live conductor of the public electric supply breaks
down and touches the earth. The consequences will be


(A) The supply voltage will drop (C) current will flow to the earth
(B) The supply voltage will increase (D) no current will flow in the conductor.
21. In the transmission system a feeder feeds power to
(A) Service mains (C) distributors
(B) Generating stations (D) all of the above.
23. The chances of corona are maximum during
(A) Summer heat (C) Dry weather
(B) Winter (D) humid weather.




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EXAMPLE: Consider the following distribution network. Distributed load along the branch lines is J = 10 W/m. A factory draws a power of 30 kW at cos φ = 0,8 lagging from pole 49. Diversity factor of the load is 70%. Copper conductor of (3 ×70 +16/50) is used between poles 49 and 61. Find the total relative voltage drop along the longest distributor line. Check whether the selected cross-section is appropriate. Permissible voltage drop is 5% along the distributor.
Redrawing the distribution network.
(5 x 10)
(3 x 70 +16/50)
(3 x 10)
30 kW
%70 div. factor
p.f. = 0,8 lagging
21 kW


Branch powers :
Sub-branch powers :
Sub-branch powers :
Maximum voltge drop along the distribution network:
%ΔV = %ΔVz + %ΔVx
%ΔV = Kr.I.P.10⁻⁷ + M.ℓ.Q.10⁻⁷
P :
Kr :
%ΔVx :
burada son hali var çizimin, bu hale gelmesi gerekiyor



Poles	%ΔVx
51-61	$12.4.240.1200.10^{-7} = 0,357$
49-51	$12.4.80.4400.10^{-7} = 0,436$
45-49	$1,77.80.26200.10^{-7} = 0,370$
34-45	$1,77.160.28200.10^{-7} = 0,798$
1-34	$1,77.160.32250.10^{-7} = 0,913$
Tr-1	$1,77.15.33050.10^{-7} = 0,0876$

Total %ΔVx = 2,96

Reactive component of voltage drop is only due to the mill at pole 49. The reactive power supplied to the mill is:

Q = (P.tgφ).(diversity factor)

Cosφ = 0,8 ⇒ φ = 36,9° , tg 36,9° = 0,75

Q = (30000.0,75).0,70 = 15750 (VARs) Inductive

Reactive voltage drop due to the mill is:

$\%\Delta V_X = M.\ell.Q.10^{-7}$

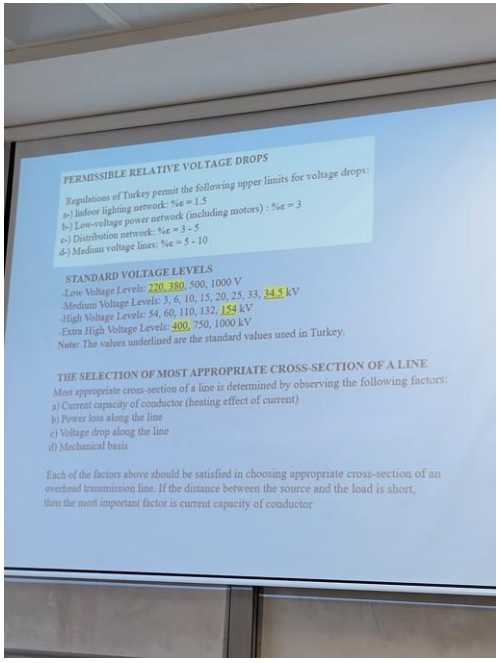
For q = 70 (mm^2), M = 1,98.

$\%\Delta V_X = 1,98.415.15750.10^{-7} = 1,294$

Total relative voltage drop (maximum relative voltage drop):

$\%\Delta V = \%\Delta V_R + \%\Delta V_X = 2,96 + 1,29 = 4,25\% < 5\%$

The conductor cross-sections used are appropriate.



PERMISSIBLE RELATIVE VOLTAGE DROPS

Regulations of Turkey permit the following upper limits for voltage drops:

a-) Indoor lighting network: $\%\varepsilon = 1.5$

b-) Low-voltage power network (including motors) : $\%\varepsilon = 3$

c-) Distribution network: $\%\varepsilon = 3 - 5$

d-) Medium voltage lines: $\%\varepsilon = 5 - 10$

STANDARD VOLTAGE LEVELS

-Low Voltage Levels: 220, 380, 500, 1000 V

-Medium Voltage Levels: 3, 6, 10, 15, 20, 25, 33, 34.5 kV

-High Voltage Levels: 54, 60, 110, 132, 154 kV

-Extra High Voltage Levels: 400, 750, 1000 kV

Note: The values underlined are the standard values used in Turkey.

THE SELECTION OF MOST APPROPRIATE CROSS-SECTION OF A LINE

Most appropriate cross-section of a line is determined by observing the following factors:

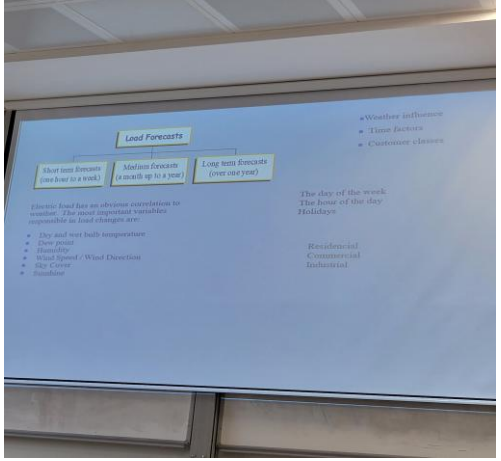
a) Current capacity of conductor (heating effect of current)

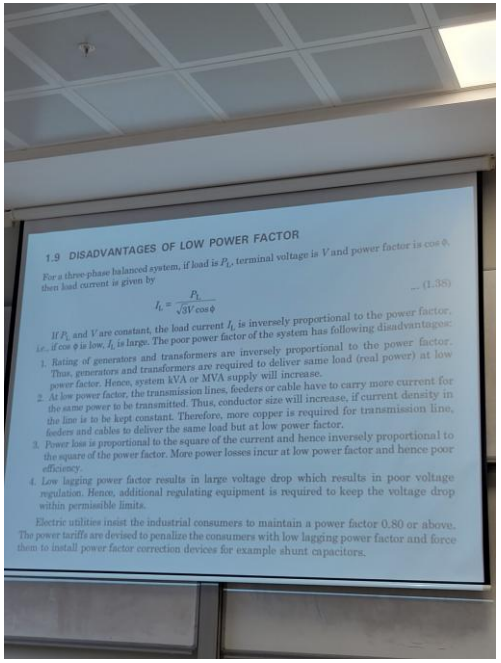
b) Power loss along the line

c) Voltage drop along the line

d) Mechanical basis

Each of the factors above should be satisfied in choosing appropriate cross-section of an overhead transmission line. If the distance between the source and the load is short, then the most important factor is current capacity of conductor





1.9 DISADVANTAGES OF LOW POWER FACTOR

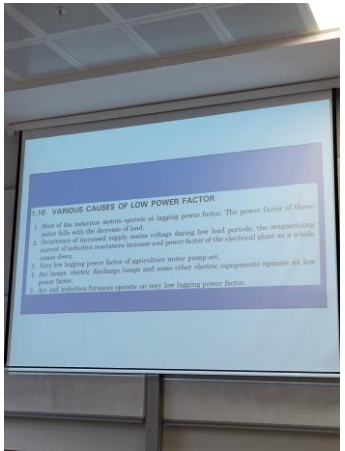
For a three-phase balanced system, if load is P_L, terminal voltage is V and power factor is $\cos\phi$, then load current is given by

$$I_L = \frac{P_L}{\sqrt{3}V\cos\phi} \quad \ldots (1.38)$$

If P_L and V are constant, the load current I_L is inversely proportional to the power factor, i.e., if $\cos\phi$ is low, I_L is large. The poor power factor of the system has following disadvantages:

1. Rating of generators and transformers are inversely proportional to the power factor. Thus, generators and transformers are required to deliver same load (real power) at low power factor. Hence, system kVA or MVA supply will increase.
2. At low power factor, the transmission lines, feeders or cable have to carry more current for the same power to be transmitted. Thus, conductor size will increase, if current density in the line is to be kept constant. Therefore, more copper is required for transmission line, feeders and cables to deliver the same load but at low power factor.
3. Power loss is proportional to the square of the current and hence inversely proportional to the square of the power factor. More power losses incur at low power factor and hence poor efficiency.
4. Low lagging power factor results in large voltage drop which results in poor voltage regulation. Hence, additional regulating equipment is required to keep the voltage drop within permissible limits.

Electric utilities insist the industrial consumers to maintain a power factor 0.80 or above. The power tariffs are devised to penalize the consumers with low lagging power factor and force them to install power factor correction devices for example shunt capacitors.



1.10 VARIOUS CAUSES OF LOW POWER FACTOR

1. Most of the induction motors operate at lagging power factor. The power factor of these motor falls with the decrease of load.
2. Occurrence of increased supply mains voltage during low load periods, the magnetizing current of inductive machinery increase and power factor of the electrical plant as a whole comes down.
3. Very low lagging power factor of agriculture motor pump-set.
4. Arc lamps, electric discharge lamps and some other electric equipments operate at low power factor.
5. Arc and induction furnaces operate on very low lagging power factor.

This is a simple series circuit. The relationship between sending-end, receiving-end voltages and currents can be written as:

$$\begin{bmatrix} V_S \\ I_S \end{bmatrix} = \begin{bmatrix} 1 & Z \\ 0 & 1 \end{bmatrix} \begin{bmatrix} V_R \\ I_R \end{bmatrix} \quad \ldots(\ .4)$$

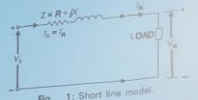


Fig. 1: Short line model.

The phasor diagram for the short-line is shown in Fig. 2 for lagging load current.

From Fig. 2, we can write

$$|V_S| \cos(\delta_S - \delta_R) = |I|R \cos\delta_R + |I| X \sin\delta_R + |V_R| \quad \ldots(\ .5)$$

$(\delta_S - \delta_R)$ is very small, $\therefore \cos(\delta_S - \delta_R) = 1.0$

$$\therefore \quad |V_S| = |V_R| + |I|\ (R \cos\delta_R + X \sin\delta_R) \quad \ldots(6.\ )$$

Equation (6) is quite accurate for the normal range of load.

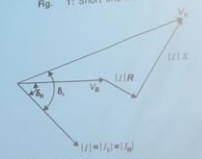


Fig. 2: Phasor diagram.

bunu ve çeşitlerini çizmeyi iyice öğren

VOLTAGE REGULATION

Voltage regulation of the transmission line may be defined as the percentage change in voltage at the receiving end of the line (expressed as percentage of full-load voltage) in going from no-load to full-load.

$$\text{Percent voltage regulation} = \frac{|V_R^{NL}| - |V_R^{FL}|}{|V_R^{FL}|} \times 100 \quad \ldots(\ .7)$$

where $|V_R^{NL}|$ = magnitude of no-load receiving end voltage

$|V_R^{FL}|$ = magnitude of full-load receiving end voltage

At no load, $I_R = 0$, $V_R = V_R^{NL}$ and from eqn. (.3),

$$V_R^{NL} = \frac{V_S}{A} \quad \ldots(\ .8)$$

Using eqns. (.7) and (.8), we get,

$$\text{Percentage Voltage regulation} = \frac{|V_S| - |A||V_R^{FL}|}{|A||V_R^{FL}|} \times 100 \quad \ldots(\ .9)$$

For a short line, $|A| = 1.0$, $|V_R^{FL}| = |V_R|$

$$\therefore \text{ Percent voltage regulation} = \frac{|V_S| - |V_R|}{|V_R|} \times 100 \quad \ldots(\ .10)$$

Using eqns. (10) and (6), we get,

MEDIUM TRANSMISSION LINE

For the lines more than 80 km long and below 250 km in length are treated as medium length lines, and the line charging current becomes appreciable and the shunt capacitance must be considered. For medium length lines, half of the shunt capacitance may be considered to be lumped at each end of the line. This is referred to as the nominal π model as shown in Fig. 3. The sending end voltage and current for the nominal π model are obtained as follows:

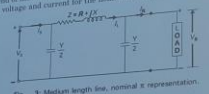


Fig. 3: Medium length line, nominal π representation.

From KCL, the current in the series impedance designated by I_L is

$$I_L = I_R + \frac{Y}{2} V_R \qquad \ldots(13)$$

From KVL, the sending end voltage is

$$V_S = V_R + ZI_L \qquad \ldots(14)$$

From eqns. (14) and (13), we get,

$$V_S = \left(1 + \frac{ZY}{2}\right) V_R + ZI_R \qquad \ldots(15)$$

The sending end current is,

$$I_S = I_L + \frac{Y}{2} V_S \qquad \ldots(16)$$

From eqns. (16), (15) and (13), we get,

$$I_S = Y\left(1 + \frac{ZY}{4}\right) V_R + \left(1 + \frac{ZY}{2}\right) I_R \qquad \ldots(17)$$

Eqns (15) and (17) can be written in matrix form.

$$\begin{bmatrix} V_S \\ I_S \end{bmatrix} = \begin{bmatrix} \left(1 + \frac{ZY}{2}\right) & Z \\ Y\left(1 + \frac{ZY}{4}\right) & \left(1 + \frac{ZY}{2}\right) \end{bmatrix} \begin{bmatrix} V_R \\ I_R \end{bmatrix} \qquad (18)$$

Therefore, the *ABCD* constants for the nominal π model are given by

$$A = \left(1 + \frac{ZY}{2}\right), \quad B = Z,$$

$$C = Y\left(1 + \frac{ZY}{4}\right), \quad D = \left(1 + \frac{ZY}{2}\right)$$

LONG TRANSMISSION LINE

For short and medium length lines, accurate models were obtained by assuming the line parameters to be lumped. In case the lines are more than 250 km long, for accurate solutions the parameters must be taken as distributed uniformly along the length as a result of which the voltages and currents will vary from point to point on the line. In this section, expressions for voltage and current at any point on the line are derived. Then, based on these equations, an equivalent π model is obtained for long transmission line. Figure 4 shows one phase of a distributed line of length *l* km.

Example 2: A 220 kV, three phase transmission line is 60 km long. The resistance is 0.15 Ω/Km and the inductance 1.4 mH/Km. Use the short line model to find the voltage and power at the sending end and the voltage regulation and efficiency when the line is supplying a three phase load of:
(a) 300 MVA at 0.8 pf lagging at 220 kV
(b) 300 MVA at 0.8 pf leading at 220 kV

Solution:

$$R = 0.15 \times 60 = 9\ \Omega$$

Assuming $f = 50$ Hz.

$$X = 2\pi \times 50 \times 1.4 \times 10^{-3} \times 60 = 26.39\ \Omega.$$

(a) Receiving end voltage per phase is

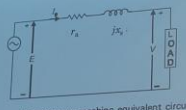
$$V_R = \frac{220\angle 0^\circ}{\sqrt{3}} = 127\angle 0^\circ \text{ kV}$$

The three phase apparent power is 300 MVA at 0.8 pf lagging

$$\therefore \quad \phi = 36.87^\circ$$

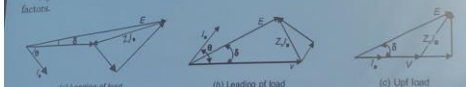
$$\therefore \quad S = 300\angle 36.87^\circ = (240 + j180)\ \text{MVA}$$

$E = V + (r_a + jx_s)I_a$



Synchronous machine equivalent circuit.

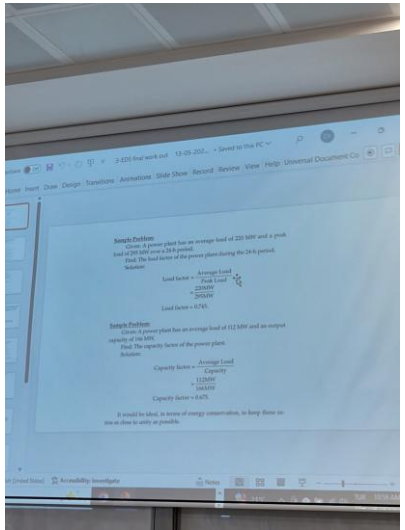
Figure shows the phasor diagram of the generator for lagging, unity and leading power factors.

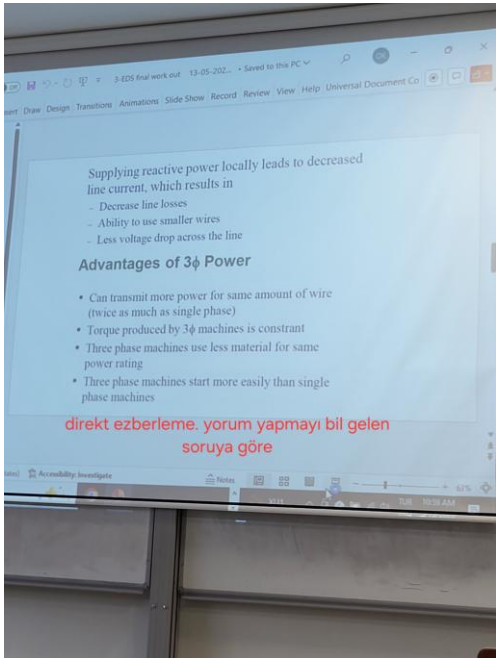


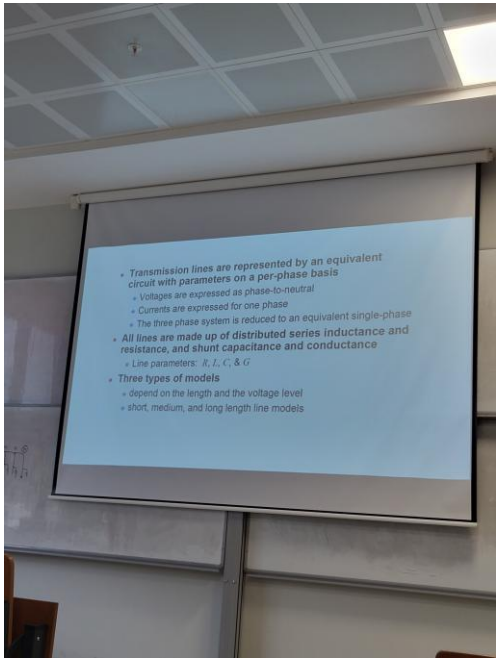
(a) Logging pf load (b) Leading pf load (c) Upf load

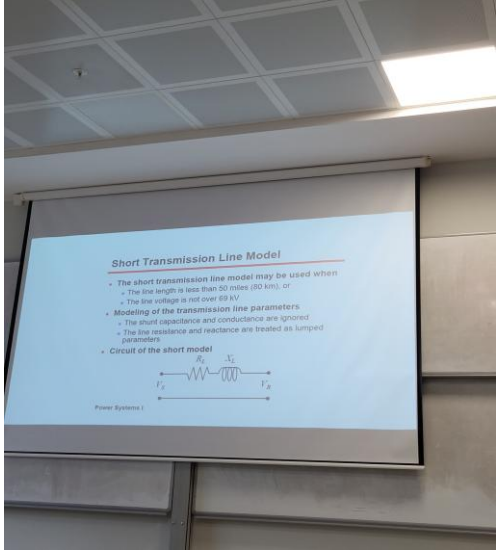
Phasor diagram of synchronous generator.

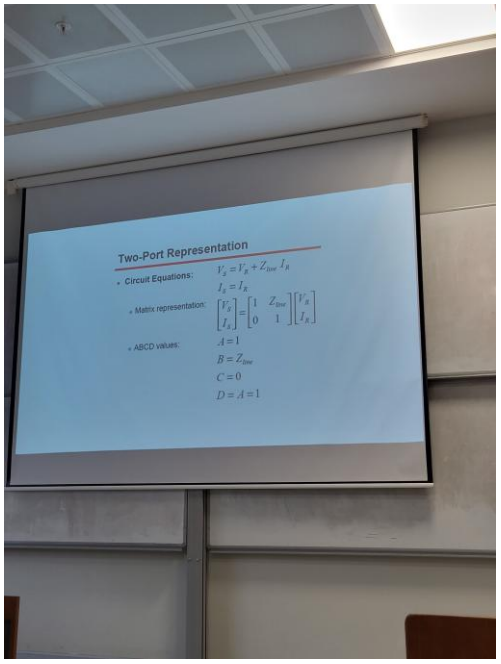
$$VR = \frac{|V_{nl}| - |V_{rated}|}{|V_{rated}|} \times 100 = \frac{|E| - |V_{rated}|}{|V_{rated}|} \times 100$$

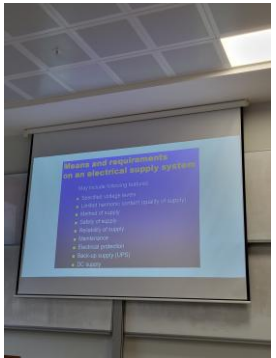


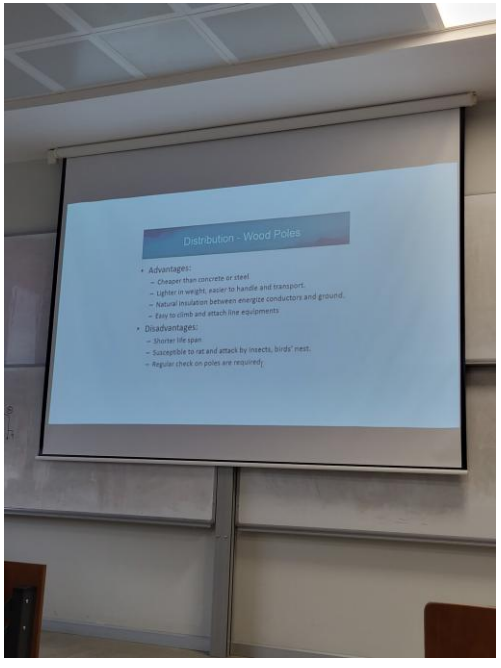

Supplying reactive power locally leads to decreased line current, which results in
– Decrease line losses
– Ability to use smaller wires
– Less voltage drop across the line
Advantages of 3ϕ Power
• Can transmit more power for same amount of wire (twice as much as single phase)
• Torque produced by 3ϕ machines is constrant
• Three phase machines use less material for same power rating
• Three phase machines start more easily than single phase machines
direkt ezberleme. yorum yapmayı bil gelen soruya göre

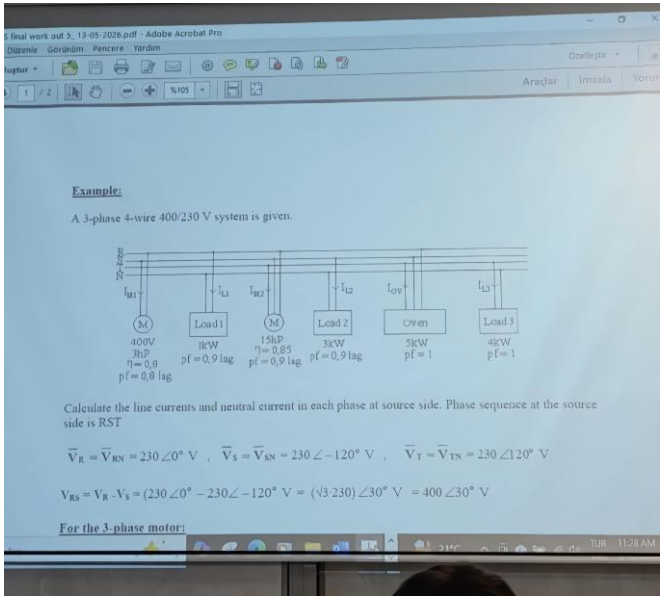

• Transmission lines are represented by an equivalent circuit with parameters on a per-phase basis
◦ Voltages are expressed as phase-to-neutral
◦ Currents are expressed for one phase
◦ The three phase system is reduced to an equivalent single-phase
• All lines are made up of distributed series inductance and resistance, and shunt capacitance and conductance
◦ Line parameters: R, L, C, & G
• Three types of models
◦ depend on the length and the voltage level
◦ short, medium, and long length line models

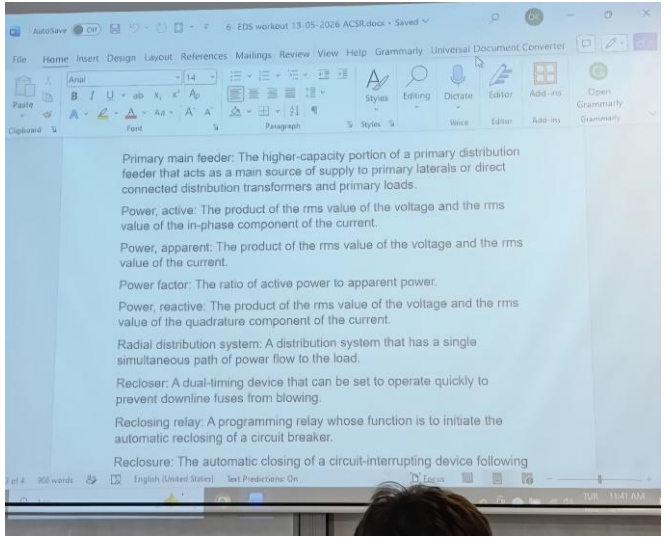

Short Transmission Line Model
The short transmission line model may be used when
The line length is less than 50 miles (80 km), or
The line voltage is not over 69 kV
Modeling of the transmission line parameters
The shunt capacitance and conductance are ignored
The line resistance and reactance are treated as lumped parameters
Circuit of the short model
R_L
X_L
V_S
V_R
Power Systems I

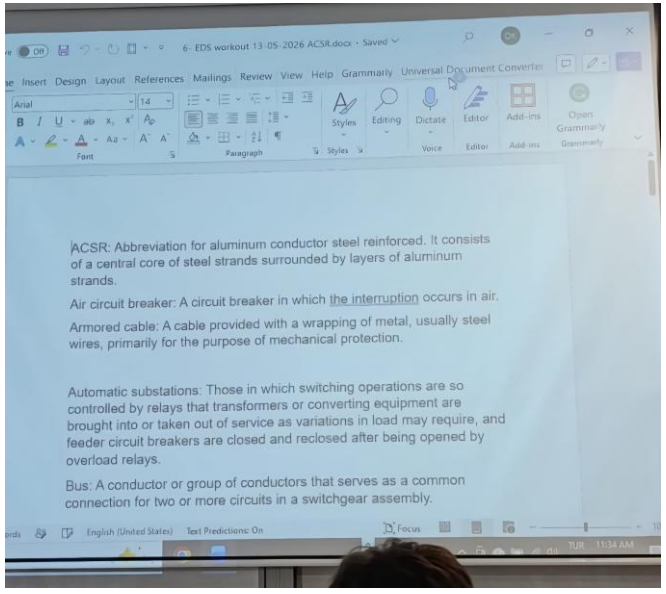

Two-Port Representation
Circuit Equations:
$V_S = V_R + Z_{line} I_R$
$I_S = I_R$
Matrix representation:
$\begin{bmatrix} V_S \\ I_S \end{bmatrix} = \begin{bmatrix} 1 & Z_{line} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} V_R \\ I_R \end{bmatrix}$
ABCD values:
$A = 1$
$B = Z_{line}$
$C = 0$
$D = A = 1$

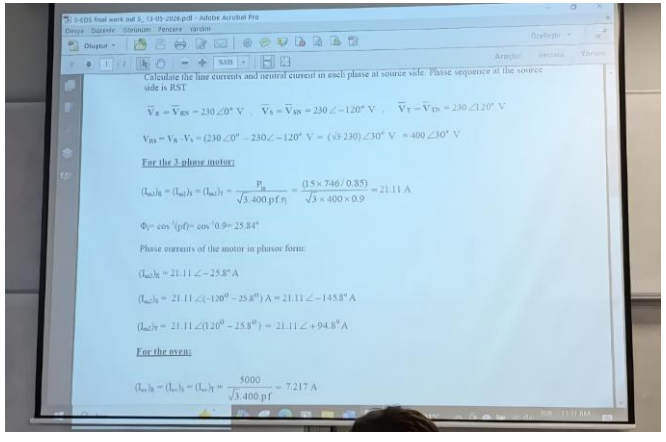

Means and requirements on an electrical supply system

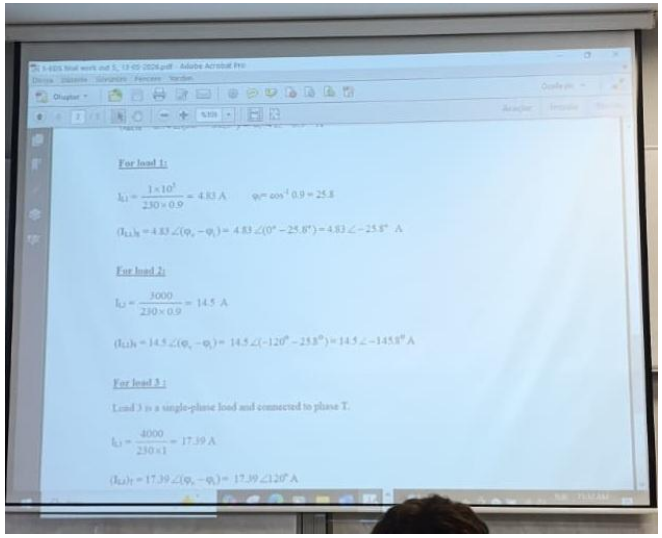

Distribution - Wood Poles
• Advantages:
– Cheaper than concrete or steel
– Lighter in weight, easier to handle and transport.
– Natural insulation between energize conductors and ground.
– Easy to climb and attach line equipments
• Disadvantages:
– Shorter life span
– Susceptible to rat and attack by insects, birds' nest.
– Regular check on poles are required.

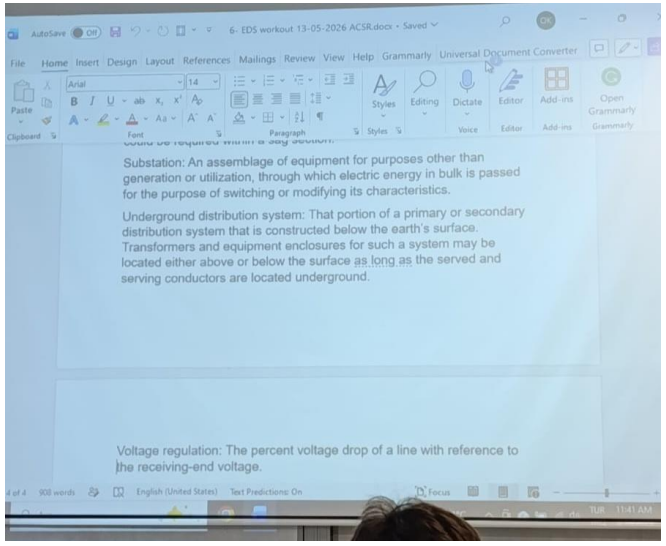

final work out 5_13-05-2026.pdf - Adobe Acrobat Pro
Düzenle Görünüm Pencere Yardım
Özellikler
Araçlar
İmzala
%105
Example:
A 3-phase 4-wire 400/230 V system is given.
230/400
Load 1
1kW
pf = 0,9 lag
15hP
η = 0,85
pf = 0,9 lag
Load 2
3kW
pf = 0,9 lag
Oven
5kW
pf = 1
Load 3
4kW
pf = 1
400V
3hP
η = 0,8
pf = 0,8 lag
Calculate the line currents and neutral current in each phase at source side. Phase sequence at the source side is RST
For the 3-phase motor:
TUR 11:28 AM

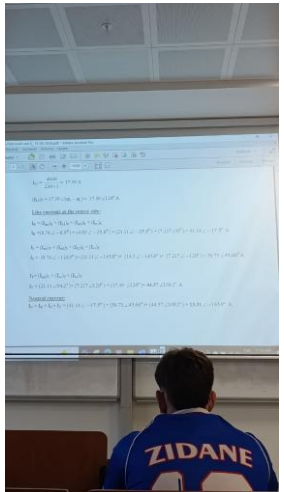

6- EDS workout 13-05-2026 ACSR.docx • Saved
Primary main feeder: The higher-capacity portion of a primary distribution feeder that acts as a main source of supply to primary laterals or direct connected distribution transformers and primary loads.
Power, active: The product of the rms value of the voltage and the rms value of the in-phase component of the current.
Power, apparent: The product of the rms value of the voltage and the rms value of the current.
Power factor: The ratio of active power to apparent power.
Power, reactive: The product of the rms value of the voltage and the rms value of the quadrature component of the current.
Radial distribution system: A distribution system that has a single simultaneous path of power flow to the load.
Recloser: A dual-timing device that can be set to operate quickly to prevent downline fuses from blowing.
Reclosing relay: A programming relay whose function is to initiate the automatic reclosing of a circuit breaker.
Reclosure: The automatic closing of a circuit-interrupting device following

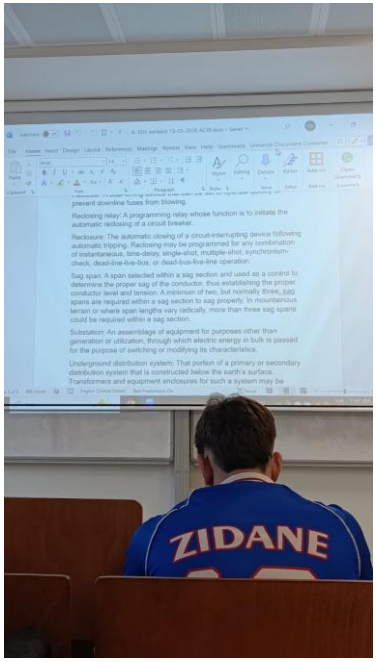

6- EDS workout 13-05-2026 ACSR.docx • Saved
ACSR: Abbreviation for aluminum conductor steel reinforced. It consists of a central core of steel strands surrounded by layers of aluminum strands.
Air circuit breaker: A circuit breaker in which the interruption occurs in air.
Armored cable: A cable provided with a wrapping of metal, usually steel wires, primarily for the purpose of mechanical protection.
Automatic substations: Those in which switching operations are so controlled by relays that transformers or converting equipment are brought into or taken out of service as variations in load may require, and feeder circuit breakers are closed and reclosed after being opened by overload relays.
Bus: A conductor or group of conductors that serves as a common connection for two or more circuits in a switchgear assembly.

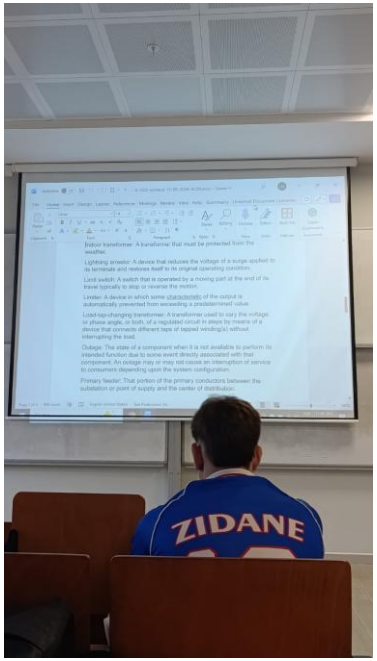

Calculate the line currents and neutral current in each phase at source side. Phase sequence at the source side is RST
For the 3-phase motor:
For the oven:
Phase currents of the motor in phasor form:

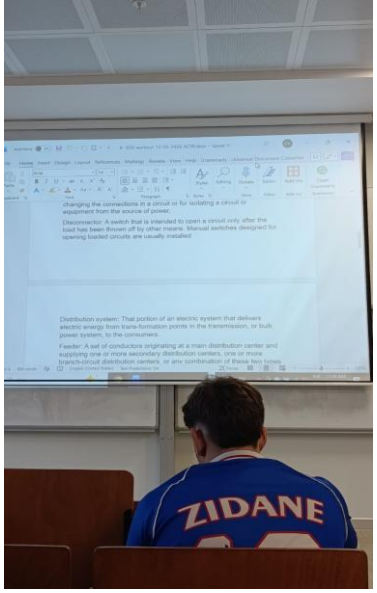

For load 1:
$I_{L1} = \frac{1\times10^3}{230\times0.9} = 4.83$ A
$\varphi_1 = \cos^{-1}0.9 = 25.8$
$(I_{L1})_R = 4.83\angle(\varphi_v - \varphi_1) = 4.83\angle(0° - 25.8°) = 4.83\angle -25.8°$ A
For load 2:
$I_{L2} = \frac{3000}{230\times0.9} = 14.5$ A
$(I_{L2})_R = 14.5\angle(\varphi_v - \varphi_2) = 14.5\angle(-120° - 25.8°) = 14.5\angle -145.8°$ A
For load 3 :
Load 3 is a single-phase load and connected to phase T.
$I_{L3} = \frac{4000}{230\times1} = 17.39$ A
$(I_{L3})_T = 17.39\angle(\varphi_v - \varphi_3) = 17.39\angle 120°$ A

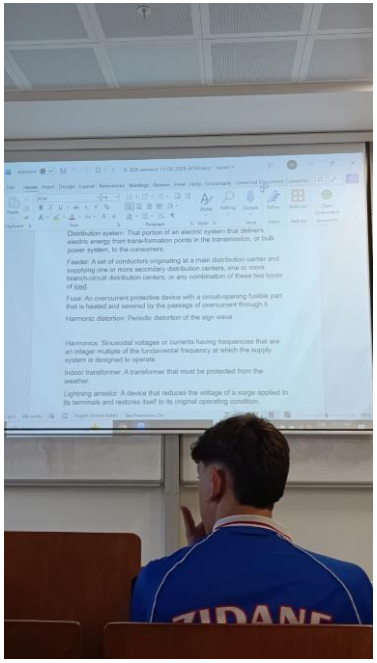

6- EDS workout 13-05-2026 ACSR.docx • Saved
Substation: An assemblage of equipment for purposes other than generation or utilization, through which electric energy in bulk is passed for the purpose of switching or modifying its characteristics.
Underground distribution system: That portion of a primary or secondary distribution system that is constructed below the earth's surface. Transformers and equipment enclosures for such a system may be located either above or below the surface as long as the served and serving conductors are located underground.
Voltage regulation: The percent voltage drop of a line with reference to the receiving-end voltage.

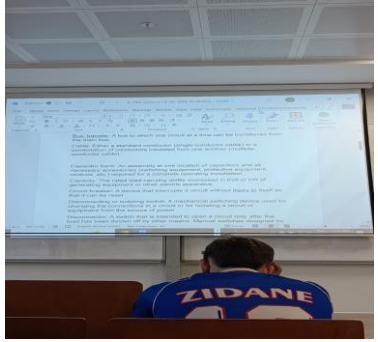

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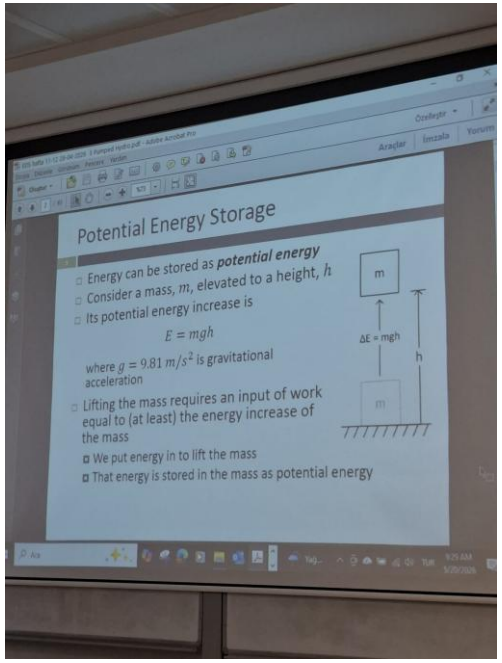

prevent downline fuses from blowing.
Reclosing relay: A programming relay whose function is to initiate the automatic reclosing of a circuit breaker.
Reclosure: The automatic closing of a circuit-interrupting device following automatic tripping. Reclosing may be programmed for any combination of instantaneous, time-delay, single-shot, multiple-shot, synchronism-check, dead-line-live-bus, or dead-bus-live-line operation.
Sag span: A span selected within a sag section and used as a control to determine the proper sag of the conductor, thus establishing the proper conductor level and tension. A minimum of two, but normally three, sag spans are required within a sag section to sag properly. In mountainous terrain or where span lengths vary radically, more than three sag spans could be required within a sag section.
Substation: An assemblage of equipment for purposes other than generation or utilization, through which electric energy in bulk is passed for the purpose of switching or modifying its characteristics.
Underground distribution system: That portion of a primary or secondary distribution system that is constructed below the earth's surface. Transformers and equipment enclosures for such a system may be
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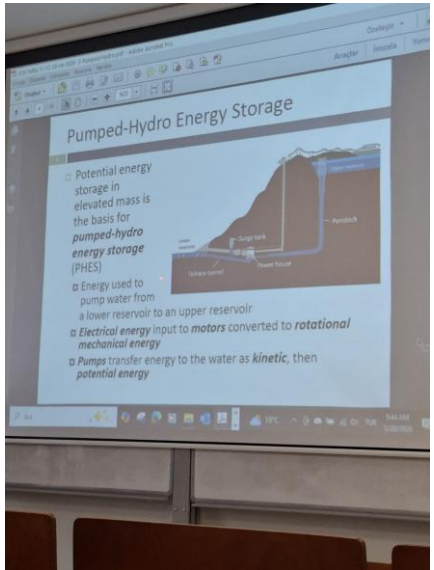

Indoor transformer: A transformer that must be protected from the weather.
Lightning arrester: A device that reduces the voltage of a surge applied to its terminals and restores itself to its original operating condition.
Limit switch: A switch that is operated by a moving part at the end of its travel typically to stop or reverse the motion.
Limiter: A device in which some characteristic of the output is automatically prevented from exceeding a predetermined value.
Load-tap-changing transformer: A transformer used to vary the voltage, or phase angle, or both, of a regulated circuit in steps by means of a device that connects different taps of tapped winding(s) without interrupting the load.
Outage: The state of a component when it is not available to perform its intended function due to some event directly associated with that component. An outage may or may not cause an interruption of service to consumers depending upon the system configuration.
Primary feeder: That portion of the primary conductors between the substation or point of supply and the center of distribution.
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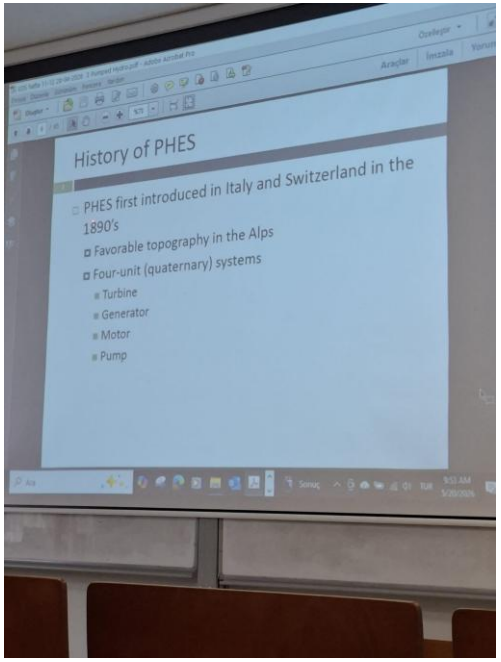

changing the connections in a circuit or for isolating a circuit or equipment from the source of power.
Disconnector: A switch that is intended to open a circuit only after the load has been thrown off by other means. Manual switches designed for opening loaded circuits are usually installed
Distribution system: That portion of an electric system that delivers electric energy from trans-formation points in the transmission, or bulk power system, to the consumers.
Feeder: A set of conductors originating at a main distribution center and supplying one or more secondary distribution centers, one or more branch-circuit distribution centers, or any combination of these two types
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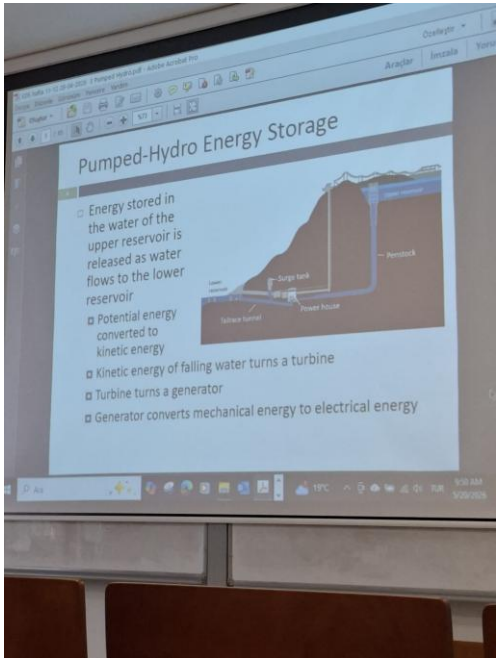

Distribution system: That portion of an electric system that delivers electric energy from trans-formation points in the transmission, or bulk power system, to the consumers.
Feeder: A set of conductors originating at a main distribution center and supplying one or more secondary distribution centers, one or more branch-circuit distribution centers, or any combination of these two types of load.
Fuse: An overcurrent protective device with a circuit-opening fusible part that is heated and severed by the passage of overcurrent through it.
Harmonic distortion: Periodic distortion of the sign wave.
Harmonics: Sinusoidal voltages or currents having frequencies that are an integer multiple of the fundamental frequency at which the supply system is designed to operate.
Indoor transformer: A transformer that must be protected from the weather.
Lightning arrestor: A device that reduces the voltage of a surge applied to its terminals and restores itself to its original operating condition.

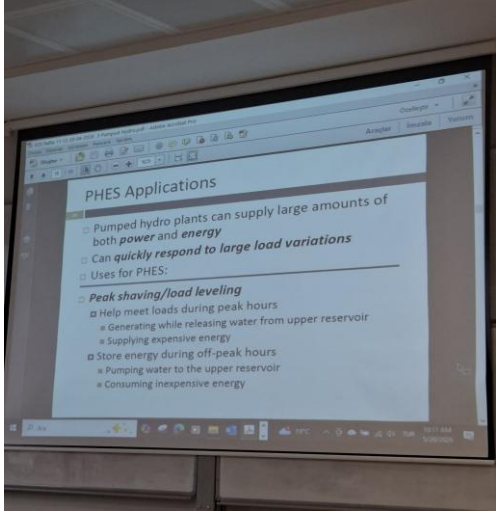

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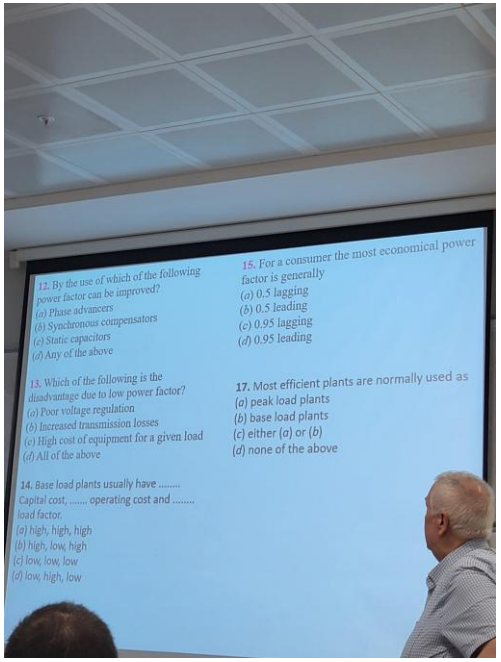

Potential Energy Storage
Energy can be stored as potential energy
Consider a mass, m, elevated to a height, h
Its potential energy increase is
E = mgh
where g = 9.81 m/s² is gravitational acceleration
Lifting the mass requires an input of work equal to (at least) the energy increase of the mass
We put energy in to lift the mass
That energy is stored in the mass as potential energy
m
ΔE = mgh
h
m

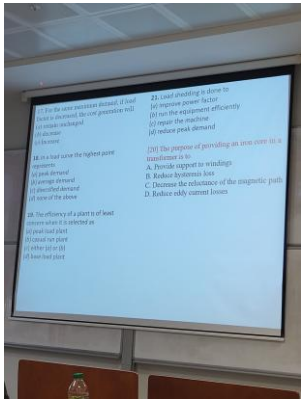

Pumped-Hydro Energy Storage
Potential energy storage in elevated mass is the basis for pumped-hydro energy storage (PHES)
Energy used to pump water from a lower reservoir to an upper reservoir
Electrical energy input to motors converted to rotational mechanical energy
Pumps transfer energy to the water as kinetic, then potential energy
Penstock
Surge tank
Power house

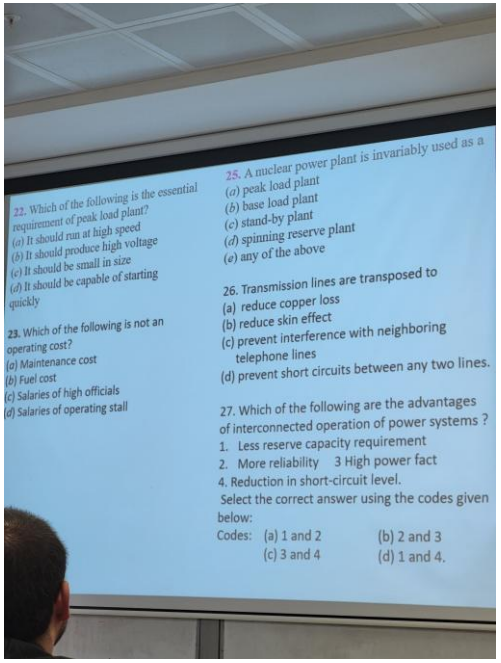

History of PHES
PHES first introduced in Italy and Switzerland in the 1890's
Favorable topography in the Alps
Four-unit (quaternary) systems
Turbine
Generator
Motor
Pump


Pumped-Hydro Energy Storage
Energy stored in the water of the upper reservoir is released as water flows to the lower reservoir
Potential energy converted to kinetic energy
Kinetic energy of falling water turns a turbine
Turbine turns a generator
Generator converts mechanical energy to electrical energy
Penstock
Surge tank
Power house
Tailrace tunnel


PHES Applications
Pumped hydro plants can supply large amounts of both power and energy
Can quickly respond to large load variations
Uses for PHES:
Peak shaving/load leveling
Help meet loads during peak hours
Generating while releasing water from upper reservoir
Supplying expensive energy
Store energy during off-peak hours
Pumping water to the upper reservoir
Consuming inexpensive energy


12. By the use of which of the following power factor can be improved?
(a) Phase advancers
(b) Synchronous compensators
(c) Static capacitors
(d) Any of the above
13. Which of the following is the disadvantage due to low power factor?
(a) Poor voltage regulation
(b) Increased transmission losses
(c) High cost of equipment for a given load
(d) All of the above
14. Base load plants usually have ..... Capital cost, ..., operating cost and ..... load factor.
(a) high, high, high
(b) high, low, high
(c) low, low, low
(d) low, high, low
15. For a consumer the most economical power factor is generally
(a) 0.5 lagging
(b) 0.5 leading
(c) 0.95 lagging
(d) 0.95 leading
17. Most efficient plants are normally used as
(a) peak load plants
(b) base load plants
(c) either (a) or (b)
(d) none of the above





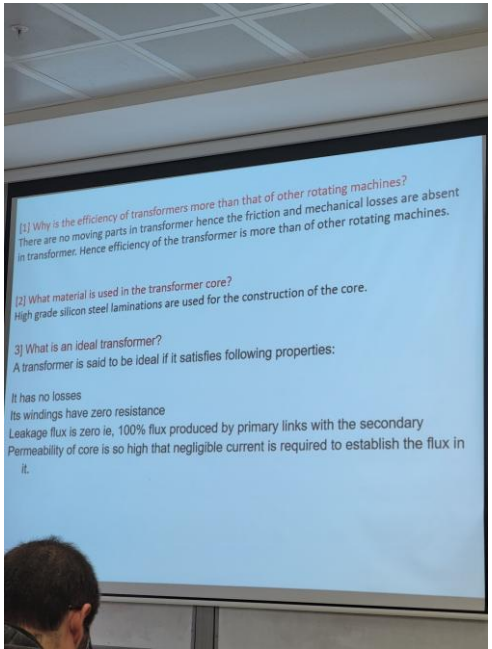
22. Which of the following is the essential requirement of peak load plant?
(a) It should run at high speed
(b) It should produce high voltage
(c) It should be small in size
(d) It should be capable of starting quickly

23. Which of the following is not an operating cost?
(a) Maintenance cost
(b) Fuel cost
(c) Salaries of high officials
(d) Salaries of operating stall

25. A nuclear power plant is invariably used as a
(a) peak load plant
(b) base load plant
(c) stand-by plant
(d) spinning reserve plant
(e) any of the above

26. Transmission lines are transposed to
(a) reduce copper loss
(b) reduce skin effect
(c) prevent interference with neighboring telephone lines
(d) prevent short circuits between any two lines.

27. Which of the following are the advantages of interconnected operation of power systems ?
1. Less reserve capacity requirement
2. More reliability 3 High power fact
4. Reduction in short-circuit level.
Select the correct answer using the codes given below:
Codes: (a) 1 and 2 (b) 2 and 3
(c) 3 and 4 (d) 1 and 4.

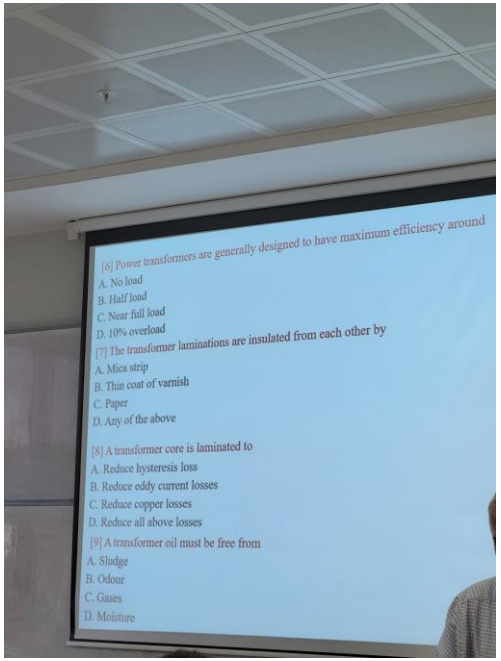


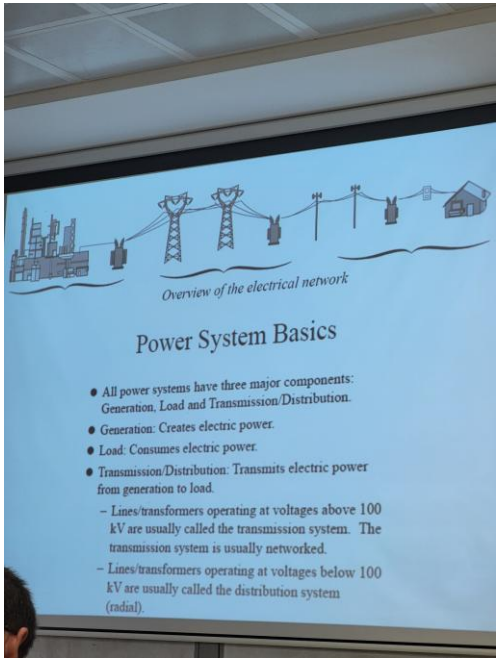
[1] Why is the efficiency of transformers more than that of other rotating machines?
There are no moving parts in transformer hence the friction and mechanical losses are absent in transformer. Hence efficiency of the transformer is more than of other rotating machines.

[2] What material is used in the transformer core?
High grade silicon steel laminations are used for the construction of the core.

3] What is an ideal transformer?
A transformer is said to be ideal if it satisfies following properties:

It has no losses
Its windings have zero resistance
Leakage flux is zero ie, 100% flux produced by primary links with the secondary
Permeability of core is so high that negligible current is required to establish the flux in it.


[6] Power transformers are generally designed to have maximum efficiency around
A. No load
B. Half load
C. Near full load
D. 10% overload
[7] The transformer laminations are insulated from each other by
A. Mica strip
B. Thin coat of varnish
C. Paper
D. Any of the above
[8] A transformer core is laminated to
A. Reduce hysteresis loss
B. Reduce eddy current losses
C. Reduce copper losses
D. Reduce all above losses
[9] A transformer oil must be free from
A. Sludge
B. Odour
C. Gases
D. Moisture


Overview of the electrical network
Power System Basics
All power systems have three major components: Generation, Load and Transmission/Distribution.
Generation: Creates electric power.
Load: Consumes electric power.
Transmission/Distribution: Transmits electric power from generation to load.
– Lines/transformers operating at voltages above 100 kV are usually called the transmission system. The transmission system is usually networked.
– Lines/transformers operating at voltages below 100 kV are usually called the distribution system (radial).